
Power-Centric Model of Risk & Disambiguating Societal Power-Dynamics with Concrete Economic and Mathematical Models

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Abstract

Power-Centric Framework of Risk. I propose a power-centric framework for understanding AI risks, arguing that catastrophic outcomes – from gradual disempowerment to authoritarian control – stem from common power-transfer mechanisms rather than purely technical failures. Using evolutionary game theory, I formalize how AI creates self-reinforcing dynamics that: (1) incentivize initial participation, (2) generate collective harm, (3) erect high exit barriers, and (4) ossify power structures.

Formalization of Power. The motivation here is being able to describe and aggregate dangerous power dynamics between populations with complicated relationships including ones that can't quite be assigned a utility function over long periods of time where people perhaps aren't acting rationally. Starting from basic models of power and continue breaking things to demonstrate the required mathematical properties a model of power must have.

1. Logic of Catastrophe

Not remotely started, need to build from scratch. Might be better to bring into scope after power.

1.1. Modeling States/Actors/Environment

$$s \in \text{States} \quad (1)$$

$$x \in X \text{ (actors = people, inanimate object, environment)} \quad (2)$$

environment, $e \in X$

(One actor, essentially representing 'everything else')⁽³⁾

$$\forall s \in S, \exists x, y.$$

$x \neq y \wedge x, y \in s$ (ie: All states have two unique actors)⁽⁴⁾

$\forall s \in S. e \in s$ (ie: Environment is in all states) ⁽⁵⁾

Logical Relationships & Progression of Claims Power Volume and Danger

- Little power $\Rightarrow$ little effects $\Rightarrow$ localized & manageable danger
- Large danger $\Rightarrow$ large power

Dynamics

- Balance of power $\wedge$ agents $\Rightarrow$ equilibrium
- Equilibrium $\neq$ no danger
- Equilibrium $\wedge$ something(?) $\Rightarrow$ no danger

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2. Sociological Definition of Power

- Dahlian/Pluralist Power (M) – A can get B to do something that B would not otherwise do (more rooted in the ability to get some boolean event to happen when they do/don't take some action.)
 - Capability (Could plausibly do it)
 - Control (Would do it, utility based)
- French and Raven
 - Reward Power – A 's perception that B has the ability to mediate rewards for him
 - Coercive Power – A 's perception that B has ability to mediate punishments for him (do something against their will)
- Luke's dimensions of power (more studying how decisions get made than any formal definitions of power.)

Combining the above, we will say:

- **Power** of A over B is the ability for A to determine the rewards/punishments of B (Independent of recognition which is dissimilar to French/Raven)

 Talk about in more detail. Just general background on power. This should perhaps come first in the paper.

3. Towards a Formal Model of Power

There are many different definitions of “power”, some of which describe slightly different concepts. This formalism doesn’t attempt to make a definition of “power” that encompasses all of these things. Instead, it aims to create a concrete single of power from one of these perspectives and either (a) Formally relate other definitions of power into this definition, (b) Demonstrate these other definitions are not coherent, or (c) Demonstrate that the aspects this fails to model are not behaviorally relevant to the questions we want to answer and would be best left to the philosophers (akin to the philosophy of quantum mechanics.)

“Thus we are not likely to produce certainly not for some considerable time to come anything like a single, consistent, coherent “Theory of Power.” We are much more likely to produce a variety of theories of limited scope, each of which employs some definition of power that is useful in the context of the particular piece of research or theory but different in important respects from the definitions of other studies. – Dahl’s Concept of power 1957”

Overall structure: Start from basic models of power and continue breaking things to demonstrate the mathematical properties a model of power must have.

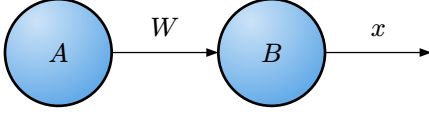
Note: State space may disappear and instead become the “behavior” of the environment which itself can be influenced via evolutionary game theory. This will come up later.

Developments in game theory have typically been motivated by being able to model a particular concept, but have also been limited by its attempt to do so. Leaving out details you don’t care about is a feature not a bug. However, it can make your models brittle when moved outside the topic.

The motivation here is being able to describe and aggregate dangerous power dynamics between populations with complicated relationships including ones that can’t quite be assigned a utility function over long periods of time where people perhaps aren’t acting rationally.

4. Generalized Dahl Framework

4.1. Defining Dahl's Power (Control)



We will start from Dahl's concept of power. However, since we define power differently, we will call Dahl's power "**control**". Instead of defining "People", they will be called "Players" to fit in with the later game theory formulations.

"Suppose I stand on a street corner and say to myself, 'I command all automobile drivers on this street to drive on the right side of the road'; suppose further that all the drivers actually do as I 'command' them to do; still, most people will regard me as mentally ill if I insist that I have enough [control] over automobile drivers to compel them to use the right side of the road. On the other hand, suppose a policeman is standing in the middle of an intersection at which most traffic ordinarily moves ahead; he orders all traffic to turn right or left; the traffic moves as he orders it to do. Then it accords with what I conceive to be the bedrock idea of [control] to say that the policeman acting in this particular role evidently has the [control] to make automobile drivers turn right or left rather than go ahead. My intuitive idea of [control], then, is something like this: A has [control] over B to the extent that he can get B to do something that B would not otherwise do. – Dahl 1957"

Amount of control (M): A's control over B with respect to the response x by means w :

$$M\left(\frac{A}{B} : W, x\right) = \Pr(B, x \mid A, w) - \Pr(B, x \mid A, \bar{w}) \quad (6)$$

$$= p_1 - p_2$$

Where $\Pr(B, x \mid A, w)$ is the probability that B takes action x given A took action w . Dahl is careful to say that this definition is not dependent on the complicated trace of causality.

Dahl's Model is made up of the following components

1. 2+ players Might not be true, each of which have Binary Action spaces (the means and the response)
2. The "connection" $\mathcal{I}$ for which these models interact. This is assumed to be unknown.
3. Probability distribution defined over:

$$\Pr : ((\text{Agent}, \text{Action}), (\text{Agent}, \text{Action})) \rightarrow [0, 1] \quad (7)$$

4. Does not require utility

4.2. Relaxing the Binary Action Space

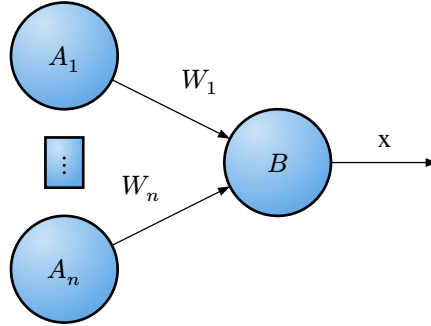
In Dahl's formalization we define the means W , an action that is (w) or is not ($\bar{w}$) taken ($W \in \{w, \bar{w}\}$). We can extend number of available actions that A has from 2-means to n-means: $W \in \mathcal{W}$ where $\mathcal{W} = \{w_1, \dots, w_n \mid n \in \mathbb{N}^+\}$:

$$M\left(\frac{A}{B} : W, x\right) = \max_{w^+ \in \mathcal{W}} \Pr(B, x \mid A, w^+) - \max_{w^- \in \mathcal{W}} \Pr(B, x \mid A, w^-) \quad (8)$$

$$= p_1 - p_2$$

In this case we are examining player A's influence over B's taking of action x . Note that if $n = 1$, representing the case where A has no ability to choose actions, then control $M = 0$.

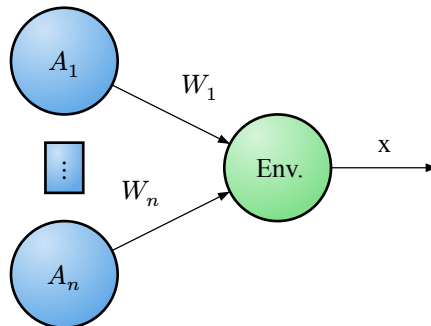
4.3. Relaxing the 2+ Player Condition



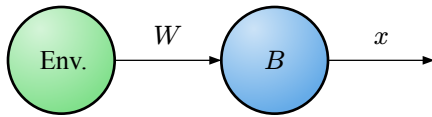
Dahl specifies how to compare control for 1+ controlling players ($A = \{A_1, \dots, A_n\}$) in the original paper. Finish. This comparative power was labeled $M_i''(A_i)$

$$M\left(\frac{A}{B} : W, x\right) = \max_{w^+ \in \mathcal{W}} \Pr(B, x \mid A, w^+) - \max_{w^- \in \mathcal{W}} \Pr(B, x \mid A, w^-) \quad (9)$$

$$= p_1 - p_2$$



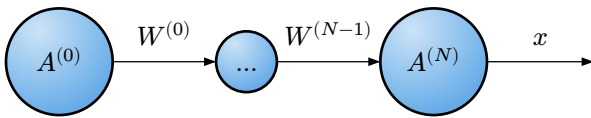
We certainly need at least one of these controlling players. However, Dahl's model of power can make due without the need for a responding player. Instead we can define control over some response from the environment instead.



You might also be able to place the Env here as A as well, although “taking actions” seems weird from an environmental perspective. But there is a useful analogue here. Kind of like how the weather controls whether you go on a picnic, in-some way you can model random decisions power on other outcomes. (Lots of issues with associating actions with the environment.)

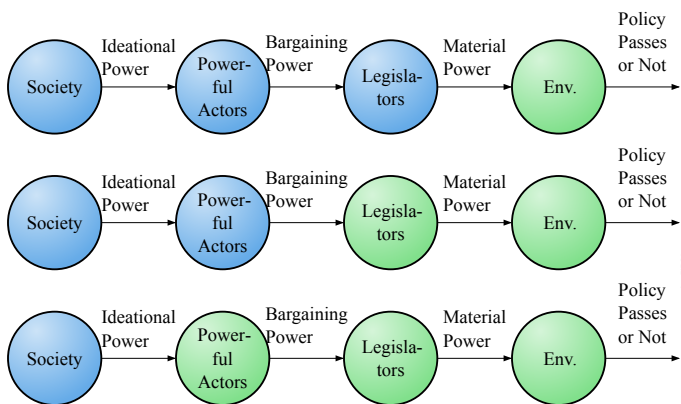
4.4. Means-Response Chains

THE BELOW DOESN'T WORK IN DAHL'S MODEL!!



Actually this seems like it might not be possible as the model does not account for the inability to choose an available action!!! Can bring in the need for a behavioral approach to control happening later on. Instantly connects with some version of game theory(?). Only works if this is the environment. Actually no, works for players as well. But takes this really weird naive stance on power that assigns more agency to a person than they actually have. This is where we can say there is a separation between power and control. (Ex: Action to burn down the entire Senate.) Best called Material power?

Also this doesn't seem be doing some kind of attribution of control properly. Capability never disappears.



Lukes criticizes Dahl and the pluralists for being too empirical, only examining the political power of observable direct decision making, whereas there are plenty of soft-power leading up to that direct decision making that play a part in both (a) Shaping the preferences of voting and policies brought to the table prior to receiving the policy (bargaining power) (b) A more cultural notion of which

ideas are even consciously considered (ideational power)

check + understand better Perhaps fun to dig up Dahl's New Haven data and demonstrate some type of this modeling. This categorization is extremely helpful for analyzing power and it will be returned to. However, can demonstrate mathematically what Luke's criticism is – ie: Doesn't really tell you who has the final say.

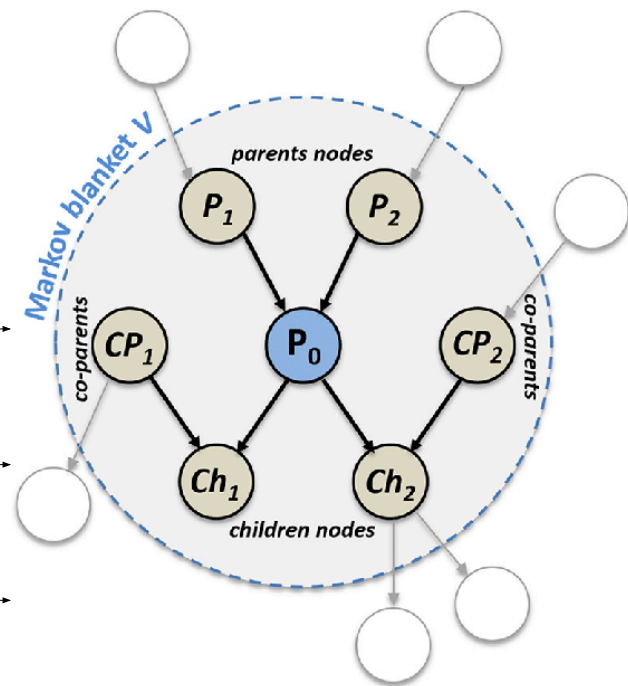
Immediately this has some connection with law, duress, consent, etc. It seems like the actual utility is an important question here, ex: no one expects positive rewards to hold up in court, but coercion does. This in some way connects with consent.

Ideological power is somehow both an information asymmetry and an alteration to one's payoff functions.

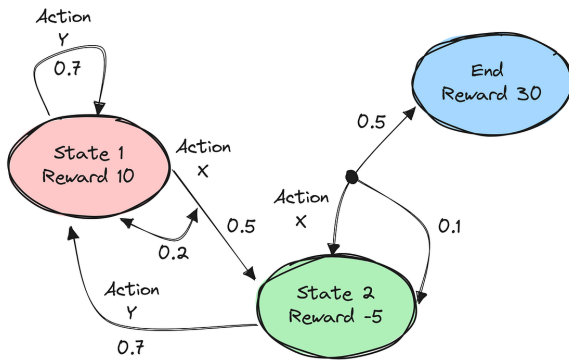
4.5. Complexity of Dahl and Need For Compact Representations

In this formalism of power, for N controlling player and a max of M actions for each, there are $N \cdot M$ action profiles $w = (w_1, \dots, w_n)$. Each of which interact through some mechanism modeled by Pr to affect the response x .

4.6. Generalizing Dahl to Markov Nets, HMMs, and MDPs



Complete.



In broad strokes, Dahl's theory of power is similar to the rich literature of markov nets in computer science literature. With a few differences:

- Taken from the perspective of actions that players take. (However these can be mapped onto priors in a markov net so long as there are no loops, which of course there are. Might not actually be an issue.)
- Control must be causal. Since this is near-impossible to prove in-full, it suffices to say that it's certainly not causal if the response happens before the action.

A few nice properties and intuitions can however be carried over:

- Markov blankets (You can isolate control by)
- Can be mapped to time-series HMMs.

Notoriously very difficult. ☐ Maybe also show some existing CS problem to apply control in a new context. ☐ Also maybe see what other studies of power exist out there that looked at causal relationships in this way. Certainly not a new understanding of power, just one that wasn't looked at broadly.

Already we are at a point where we can define control in MDPs, one of the most fundamental structures in reinforcement learning and applicable to many scenarios. ☐ Can we apply this to some textbook MDP scenario? ☐ Not sure I should transition to MDPs without bringing in utility as a pre-requisite of power.

4.7. Connecting Dahl to Game Theory

What Dahl's theory of power does not say is HOW the players in this scenario determine their actions, keeping the probability function $\Pr$ doing a lot of work here.

4.8. Criticisms of Dahl

This is not to overly simplify these models, only to ground them. Much more complex relationship with loops going back on itself over time.

5. Why Actions are Necessary for Power Analysis

5.1. The Inseparability Thesis

Claim. Power is meaningless without considering available actions – these elements are fundamentally inseparable in any coherent power framework.

5.1.1. Behavioral Power

Insight 1. Power is a function of available actions.

Insight 2. Power is a function of optimal actions assuming rationality.

I might name this control.

 **Finish.**

6. Why Absolute Power Fails to Model Power Dynamics

 Should this only be possible once utility is introduced?

 From cooperative game theory, we have weighted graphs, rule-based representations, and weighted-voting games.

6.1. The Modeling Asymmetry Thesis

Claim. Models of relative power ($\mathcal{R}$) form a proper superset of models of absolute power ($\mathcal{A}$), establishing that $\mathcal{R}$ is more expressive than $\mathcal{A}$ in modeling capacity.

Dahl says that there must be a “connection” for there to be a power dynamic.

6.1.1. Formal Statement

Definition (Power Model Spaces). Let:

- $\mathcal{A} = \{M :$
 $M \text{ models power as properties of agents}\}$
- $\mathcal{R} = \{M :$
 $M \text{ models power as relational}$
 $\text{properties between agents}\}$

Then: $\mathcal{A} \subset \mathcal{R}$ (proper subset relation)

6.1.2. Supporting Arguments

Theorem (Reduction Principle). Any absolute power model can be reformulated as a relative power model by introducing an environmental baseline.

Formal Construction. Given absolute power $P_{\text{abs}} : \text{Agent} \rightarrow \mathbb{R}^+$, we can define:

$$P_{\text{rel}(a,e)} = P_{\text{abs}(a)} - P_{\text{abs}(e)} \quad (10)$$

where $P_{\text{abs}(e)}$ serves as a zero-point reference for environment e .

 **Replace with an actual proof from graph theory**

Physical Analogy. This construction mirrors potential energy in physics, where absolute potential V is meaningless without a reference frame:

$$V_{\text{rel}} = V_{\text{abs}} - V_{\text{ground}} \quad (11)$$

Incompleteness of Absolute Power Models ($\mathcal{A} \subsetneq \mathcal{R}$).

Theorem (Limitation Theorem). Absolute power models cannot capture essential relational dynamics without introducing factors outside their framework.

 Demonstrate some examples like in my original, such as distance as one factor and just other seemingly complex set of environmental interaction factors that seem almost impossible to resolve absolute powers between entities to relative powers in a sensible way. In physics potential can be mapped to potential energy using distance, unlikely we have anything nearly as similar.

Core Problem. The power relationship between agents A and B cannot be logically derived from their absolute powers alone. Power lacks transitivity – if A dominates B and B dominates C , this does not guarantee A dominates C . This non-transitivity demonstrates that power cannot be reduced to a single absolute measure.

Example. Consider a circular dominance structure (like rock-paper-scissors):

- Rock defeats Scissors
- Scissors defeats Paper
- Paper defeats Rock

No absolute power assignment P_{abs} can capture these relationships.

Conclusion. Since capturing these dynamics requires factors $\notin \mathcal{A}$, we have $\mathcal{A} \neq \mathcal{R}$.

 **improve**

6.1.3. Proof of Proper Subset Relation

From §1.2.1: $\mathcal{A} \subseteq \mathcal{R}$

From §1.2.2: $\exists M \in \mathcal{R}$ such that $M \notin \mathcal{A}$

Therefore: $\mathcal{A} \subset \mathcal{R}$ (proper subset) $\square$

6.1.4. Power Over Environment as Logical Encapsulation of What is Meant By “Absolute Power”

📄 fin

6.1.5. Conclusion

Absolute power alone is insufficient for modeling power. For the remainder of this work, we consider only relational power – power over the environment or power over other agents.

Note: This does NOT tell us that relative power is sufficient, only that it may be.

7. Why Utility is Necessary for Power Analysis

7.1. The Inseparability Thesis

Claim. Power is meaningless without considering agent utilities – these elements are fundamentally inseparable in any coherent power framework.

NOTE: Will likely replace this utility statement.

7.1.1. Utility Alignment

Core Insight. Power is an incoherent concept under complete alignment of utilities.

Definition (Utility Alignment). Let:

- $U_A, U_B : \mathcal{S} \rightarrow \mathbb{R}$ be utility functions for agents A and B
- $\mathcal{S}$ be the state space
- $P(A \rightarrow B)$ be the power of A over B

If $U_A(s) \propto U_B(s) \forall s \in \mathcal{S}$ (perfect alignment), then $P(A \rightarrow B)$ is behaviorally irrelevant.

Illustrative Example. Consider a human-loving agent who can travel to an island, harm anywhere from 1 to 1,000,000 innocent people, and then escape unnoticed. Does their power increase with the number they *could* harm? This capacity is irrelevant:

- The human-loving agent doesn't benefit from causing harm and would prefer not to
- The potential victims have no desire to be harmed

The mere existence of this action is meaningless without considering whether it would ever be exercised.

 Replace with better example.

 Include behavioral misalignment, ie: power over

7.1.2. Agent Boundary Definition

Theorem (Aggregation Principle). Alignment relationships determine when individual power can be meaningfully aggregated into a single coherent agent:

$$\left[P_{\text{collective}} = \sum_i P_i \right] \Leftrightarrow [U_i \approx U_j \forall i, j] \quad (12)$$

Application. Humanity's collective power is discussable precisely because humans share sufficient utility alignment over key domains.

Connection to Sociology This is akin to *power-within* discussed in sociological literature.

7.1.3. Environmental Mediation

Extended Framework. When modeling multi-agent systems with environmental interactions:

- Environment E can be treated as an agent with its own dynamics
- Even without direct conflict between agents, they may compete for environmental resources
- Agents may inflict harm indirectly through environmental manipulation

Power becomes relevant when agents compete for control over shared environmental factors, even if their ultimate goals don't directly conflict.

 Also could say that their actions are some mixed strategy over a number of actions. Not sure if this is true but not a crucial point.

7.1.4. Agents in Shared State Space

Agents of course, are themselves components of state spaces and they may have preferences over the state of themselves and others. Often times they have preferences for their survival which must be modeled in the utility function as well.

7.1.5. Critique of Light-Cone Formulation

Orthogonality Scenario. Consider alien civilization Ω with utilities orthogonal to humanity H . Imagine Ω cares only about arranging neutrinos into smiley faces – an activity that has zero impact on human flourishing. Meanwhile, humans optimize matter and energy arrangements with no effect on neutrino patterns. (We can additionally imagine that neither agent meaningfully interferes with the other, perhaps humanity and these aliens never interact)

Even if Ω achieves perfect control over neutrino arrangements while humans have only partial control over matter/energy, it's meaningless to compare their “power over the environment.” The concept of environmental control is incoherent without specifying *which aspects* of the environment matter.

Key Insight. Power must be defined over relevant state space – control over factors that actually matter to the agents involved:

$$P_{\text{relevant}(A)} = \int_{\mathcal{S}_A} |\nabla U_A(s)| \, ds \quad (13)$$

where $\mathcal{S}_A := \{s \in \mathcal{S} : \partial U_A / \partial s \neq 0\}$ represents the relevant state space.

Note that this relevant state space is inherently chosen in reference to some agent and thus is incoherent outside of some utility reference frame.

7.2. Synthesis

Theorem (Completeness Requirement). A complete theory of power requires (although may be insufficient):

1. Relational framework ($\mathcal{R} \supset \mathcal{A}$)
2. Behavioral specifications for all agents (possibly without explicit utilities)
3. Action space definitions $\mathcal{A}_i$
4. Relevance constraints on state space $\mathcal{S}_i \subset \mathcal{S}$

Implication. Power analysis without game-theoretic foundations is fundamentally incomplete. $\square$

 Power of signaling. US vs Japan and communicating the existence of the nuclear bomb. US in some way doesn't have power over Japan if it doesn't benefit from either (A) Annihilating it or (B) Communicating its Annihilation.

References